

ÚLOHA 3.2

(a) $\vec{v}_1 = 1, \vec{v}_2 = x, \vec{v}_3 = x^2$

1. vektor: $\vec{w}_1 = \vec{v}_1 = 1, \|\vec{w}_1\|^2 = \int_1^5 1^2 dx = 4, \|\vec{w}_1\| = 2, \vec{e}_1 = \frac{1}{2}$

2. vektor: $\vec{w}_2 = \vec{v}_2 - \langle \vec{v}_2, \vec{e}_1 \rangle \vec{e}_1, \langle x, \frac{1}{2} \rangle = \int_1^5 x \cdot \frac{1}{2} dx = \frac{1}{2} \int_1^5 x dx = \frac{1}{2} \cdot 12 = 6, \vec{w}_2 = x - 6 \cdot \frac{1}{2} = x - 3, \|\vec{w}_2\|^2 = \int_1^5 (x-3)^2 dx = \frac{16}{3}, \|\vec{w}_2\| = \frac{4}{\sqrt{3}}, \vec{e}_2 = \frac{\sqrt{3}}{4} (x-3)$

3. vektor: $\vec{w}_3 = \vec{v}_3 - \langle \vec{v}_3, \vec{e}_1 \rangle \vec{e}_1 - \langle \vec{v}_3, \vec{e}_2 \rangle \vec{e}_2$

$$\langle x^2, \vec{e}_1 \rangle = \frac{1}{2} \int_1^5 x^2 dx = \frac{1}{2} \cdot \frac{124}{3} = \frac{62}{3} \rightarrow \frac{62}{3} \cdot \frac{1}{2} = \frac{31}{3}$$

$$\langle x^2, \vec{e}_2 \rangle = \frac{\sqrt{3}}{4} \int_1^5 x^2 (x-3) dx = 8\sqrt{3} \rightarrow 8\sqrt{3} \cdot \frac{\sqrt{3}}{4} (x-3) = 6(x-3)$$

$$\vec{w}_3 = x^2 - \frac{31}{3} - 6(x-3) = x^2 - 6x + \frac{23}{3}$$

$$\|\vec{w}_3\|^2 = \int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx = \frac{256}{45}, \|\vec{w}_3\| = \frac{16\sqrt{5}}{15} = \frac{16}{3\sqrt{5}}, \vec{e}_3 = \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$$

(b) ortog. projekce $p(x)$ funkce $\sin(x)$: $p(x) = \sum_{i=1}^3 \langle \sin x, \vec{e}_i \rangle \vec{e}_i$

$$\langle \sin x, \vec{e}_1 \rangle = \frac{1}{2} \int_1^5 \sin x dx = \frac{1}{2} (\cos(1) - \cos(5)) \approx 0,128$$

$$\langle \sin x, \vec{e}_2 \rangle = \frac{\sqrt{3}}{4} \int_1^5 \sin x (x-3) dx \approx -1,493$$

$$\langle \sin x, \vec{e}_3 \rangle = \frac{3\sqrt{5}}{16} \int_1^5 \sin x (x^2 - 6x + \frac{23}{3}) dx \approx -0,125$$

po dosazení dostaneme: $p(x) = 0,128 \cdot \frac{1}{2} - 1,493 \cdot \frac{\sqrt{3}}{4} (x-3) - 0,125 \cdot \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \approx -0,052x^2 - 0,333x + 1,604$

(c) $\|\sin x - p(x)\| = \sqrt{\int_1^5 (\sin x - p(x))^2 dx} = 0,3189$

$$\int_1^5 (\sin(x) - (-0,052x^2 - 0,333x + 1,604))^2 dx = 0,10169$$